THE HUMAN BATTERY™ PROJECT

# BEGINNER

## NINETY-DAY PROTOCOL

*You are starting from a stopped engine. One thing a month. By day 90 you will be doing more than you thought possible and it will never have been hard on any single day.*

THE MODEL

## Trillions of batteries, keeping time together

Your body is not one battery. It is a coordinated network of trillions of cellular batteries. Every cell holds a voltage across a membrane, and different tissues are built to hold different voltages. A neuron, a muscle cell and an immune cell are supposed to be different. The goal is the voltage appropriate to each tissue, never the maximum.

**Coherence** is the ability of cells to hold their proper electrical identity while coordinating their voltage, metabolism, timing and behavior with the tissue around them and with the whole organism. A cell can be individually healthy and still be out of step with its neighbours. That is what the ninety days are for.

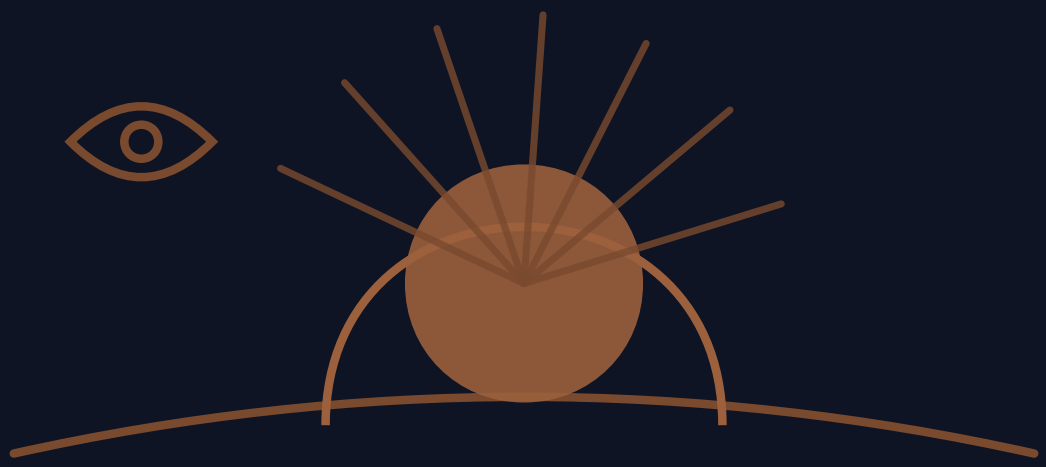
Light does two jobs. The big one is timing. Morning light tells the body what hour it is, and that signal sets the nervous system, hormones, metabolism and mitochondrial activity. The small one is chemistry in your skin: sunlight makes vitamin D and releases nitric oxide, which widens your arteries. So light is a real energy input, and a small one. Your skin takes in a lot of sunlight and nearly all of it becomes heat, because the body cannot turn light into fuel.

Eating with the season is in the protocol because different foods give your body different fuel, and daylength sets how your body handles fuel. Lining those up is the idea being tested. It is not a settled fact, and food does not carry any message about the month it grew in.

The circadian protocol is how coherence is supported. The daily log is how it is tracked. Every element in this document carries a note saying which of the four questions it answers, and where an element answers none of them, it says so.

# Circadian protocol

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.



LIGHT SETS THE CLOCK

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01 / THE CLOCK

# Circadian protocol

BEGINNER

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.

Your body runs on a clock, and the clock is set by light. Morning light tells every cell what time it is. Darkness at night lets the repair cycle run. When the clock is right, food is handled correctly, hormones release on schedule, and sleep does its job. When the clock is wrong, nothing downstream works, no matter how well you eat or train.

The sequence, every day:

LIGHT → WATER → MOVEMENT → FOOD → MOVEMENT → LIGHT → DARKNESS

### Go outside as soon as you wake up

**What to do:** Outside within 30 minutes of waking up. Fifteen minutes minimum. Sit in a chair if you need to. Shoes and socks off, bare feet on grass, dirt or sand if you have it. Face toward the sun with your eyes open. No sunglasses. Not through a window, not from the car. Outside. Do this before coffee, before food, before your phone. This is the most important fifteen minutes of your entire day.

**Why:** Your body has a clock inside it. The clock does not know what time it is until it sees the sun. When morning light hits your eyes, it tells every cell in your body that the day has started. That one signal sets up your energy, your hunger, your hormones and your sleep for the whole day. If you skip it, your body runs on the wrong time all day, like a clock that was never set.

Barefoot if you can. This is part of the morning light block, not separate from it. We include it because it costs nothing, and because standing on grass keeps people outside for the full time when shoes and a doormat do not. We are not measuring it, and we are not claiming it does anything on its own.

> **Answers Q3, Q4.** Sets the circadian clock every tissue runs on. Sleep timing and light exposure are logged daily.

### Get sun on your skin in the afternoon

**What to do:** Fifteen minutes outside in the afternoon, somewhere between noon and 3pm. Short sleeves if the weather allows. Sit or walk, either is fine. If your skin is fair, start with ten minutes and build to fifteen. Never burn.

**Why:** Morning light sets your clock. Midday sun holds it there. It is also the only time of day your skin can make vitamin D, and sun on bare skin widens your blood vessels, which lowers blood pressure. We measure both at day 0 and day 90.

> **Answers Q3, Q4.** Holds the clock set. Vitamin D is on the panel and blood pressure is measured at both ends.

### Watch the sun go down

**What to do:** Fifteen minutes outside as the sun is going down. Look toward the sunset, not straight at the sun. This is the third outdoor block of your day, and it is the one that fixes your sleep.

**Why:** The light at sunset is red and orange. That color tells your body the day is ending and it is time to start making melatonin, the hormone that makes you sleepy. Seeing the sunset is how your body knows to begin winding down. Skip it and your body does not get the message, and then you are wide awake at midnight wondering why.

> **Answers Q3.** Red light starts melatonin onset, the handover from day signaling to night signaling.

### Put on your glasses when the sun goes down

**What to do:** The moment the sun goes down and you come inside, put on your amber glasses. Wear them until you are in bed. Every screen you use at night gets a blue light filter: any color-temperature shifter on the computer, the free ones work, night mode on your phone, and a blue-blocking film on the TV. Swap the bulb in your bedroom lamp for a warm or red one this week. Add the living room next month.

**Why:** After sunset, your body expects darkness. Light bulbs and screens give off a blue light that looks like daytime to your brain. Your brain sees it and stops making melatonin, and that is why you cannot fall asleep even when you are tired. Amber or red glasses block the blue. So do warm bulbs and screen filters. It is not about the brightness. It is about the color.

> **Answers Q3.** Protects that melatonin signal from blue light that would tell the body it is still daytime.

### Make the bedroom dark and quiet

**What to do:** Your phone sleeps in another room. Starting tonight. Charge it in the kitchen. If you use it as an alarm, buy a ten-dollar alarm clock. Cover any small lights in the room, like the glow on a charger or a TV. As dark as you can get it.

**Why:** Your body does its repair work while you sleep, and it can only do it in the dark. Even a small light, like a phone screen or a charger glow, can be enough to keep the repair from starting. The phone also keeps your brain on alert, waiting for a buzz. Out of the room means your body can finally switch off.

> **Answers Q3, Q4.** Darkness is when repair signaling runs. Sleep timing and regularity are logged daily.

### Give your body hours with no food

**What to do:** Month one: nothing to eat for three hours before bed. That is the only rule. Month two: eat between 9am and 7pm, and close the window thirty minutes earlier every two weeks. By day 90 you are eating between 9am and 6pm, and you will not have noticed it happening.

**Why:** Every time you eat, your body has to stop what it is doing and deal with the food. If you eat from the minute you wake up until right before bed, your body never gets a break to do repairs. Hours without food, especially overnight, is when the cleanup happens. Late eating is the worst, because your body is trying to wind down and you just handed it a job.

> **Answers Q1, Q4.** Metabolic flexibility and substrate switching. Fasting glucose, insulin and HbA1c are on the panel.

> **The rule that matters most:** light before food, every single day. Do not eat until you have been outside.

02 / WATER

# Water and minerals

The morning glass first. Then three liters, every bottle with minerals in it.



MINERALS, LEMON, SALT

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02 / WATER

# Water and minerals

BEGINNER

## The morning glass

Before coffee, before food, before anything: 16 to 24 oz of mineral water with a pinch of unrefined mineral sea salt and 1 to 2 oz of pure organic lemon juice. Empty stomach. It is the first thing the body receives after light.

## Through the day

Three liters. It will feel like a lot at first. Carry a bottle and refill it three times. Every bottle gets a pinch of unrefined mineral sea salt, because water without minerals passes through and takes minerals with it. Most of it before your last meal.

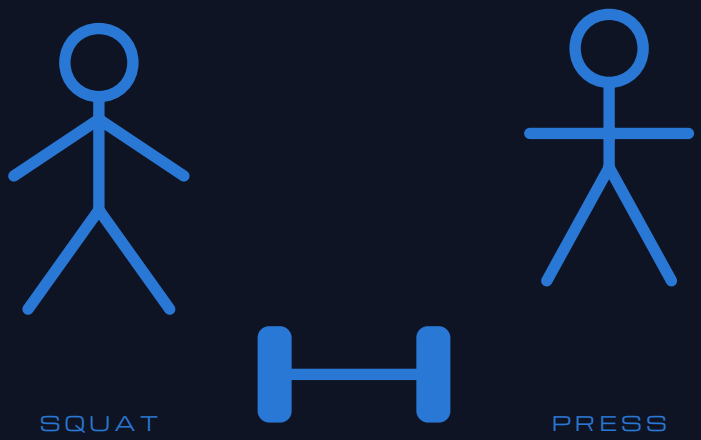
## What to drink

Bottled water with a measurable mineral content, not distilled. At home, reverse osmosis with unrefined mineral sea salt added back, a quarter teaspoon per liter, and the same salt for cooking. Nothing else. We take no commission on anything named in this guide.

> **Answers Q2, Q4.** Minerals are the ions the pumps move. Sodium, potassium and RBC magnesium are on the panel.

# Movement

Four patterns. Walks after every meal. Zone 2 for the mitochondria, intervals for the ceiling.




SQUAT, PUSH, PULL, HINGE

3

03 / MOVEMENT

# Movement

BEGINNER

### Month one

Nothing. Light, water and sleep only. You are building the clock first.

### Month two

**Walking:** ten minutes after dinner every night. After two weeks, after lunch too. After four, after every meal. **Resistance, twice a week, ten minutes:** sit-to-stand from a chair, wall push-ups, seated band rows, hip hinge with a broomstick. Three sets of what you can do. **Steps:** your current average plus a thousand.

### Month three

**Resistance, three days.** Sit-to-stand becomes bodyweight squat. Wall push-up becomes incline push-up on a counter. Band row becomes inverted row under a table. Broomstick hinge becomes kettlebell deadlift. **Walking:** twenty to thirty minutes after dinner, some of it brisk enough that talking is hard.

**No intervals before day 60, and only with physician clearance.**

> **Answers Q1, Q4.** Builds mitochondrial number and respiratory capacity. VO2max, lactate and grip strength are measured at both ends.

# Food

Protein first, fat for fuel, inside a window that closes early. The full list is in the Dietary Guidelines.



PROTEIN, FAT, THEN CARBS

4

04 / FOOD

# Food

BEGINNER

### Month one

No food rules except one: nothing three hours before bed. If you are eating fast food, you are eating fast food this month. We are building the clock first.

### Month two: the window, and one food a week

Eating window 9am to 7pm, tightening thirty minutes on the evening side every two weeks. **No snacks.** Three meals. This one rule does more than any other.

**Week 5:** eggs every morning. **Week 6:** greens with lunch. **Week 7:** sardines or salmon three times a week. **Week 8:** sauerkraut or kimchi with dinner.

### Month three: remove two things

Gluten and seed oils, starting week 9. That is the only elimination in this tier. Nightshades, grains and legumes stay in.

### Protein

Work toward 1.6 grams per kilogram by day 90. Protein first at every meal.

> **Answers Q1, Q2, Q4.** Fuel handling, plus the fatty acids and protein the membranes are built from. Glucose, insulin, HbA1c, Omega-3 Index and albumin are on the panel.

The full approved food list, the daily non-negotiables and how to build a plate are in the Dietary Guidelines document.

# Heat and cold

Stress that builds capacity. Dosed to your tier, timed away from your lifts.



HEAT COLD

STRESS THAT BUILDS CAPACITY

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# Heat and cold

BEGINNER

### Heat, twice a week

Eight to ten minutes, up to 80C (176F). Only with physician clearance if anything in the list below applies to you. Gym sauna is fine.

### Cold, three to five days a week

End your shower cold. Thirty seconds to start, building to sixty. Full cold, not cool. Breathe slowly through your nose. That is the whole cold protocol at this tier: no plunge.

> **Cold is never within six hours after strength training.** Cold before a lift is fine. Cold after a lift blunts what the lift was for. If you train in the morning, cold goes in the evening or on a day you did not lift.

> **The sauna finishes one to two hours before bed.** Later than that and the heat is still leaving your body when you are trying to fall asleep.

> **Two limits that never move.** No single cold exposure over five minutes, and no water colder than 7C (45F). These hold at every tier, and they hold when your intensity rises between cycles. Colder and longer is not the direction this goes.

> **No alcohol before or during the sauna.** Mineral water after.

> **Never plunge alone in open water. Never put your head under, and never hold your breath.** The gasp reflex is involuntary and it is what makes cold water drown people who can swim.

### Who skips this

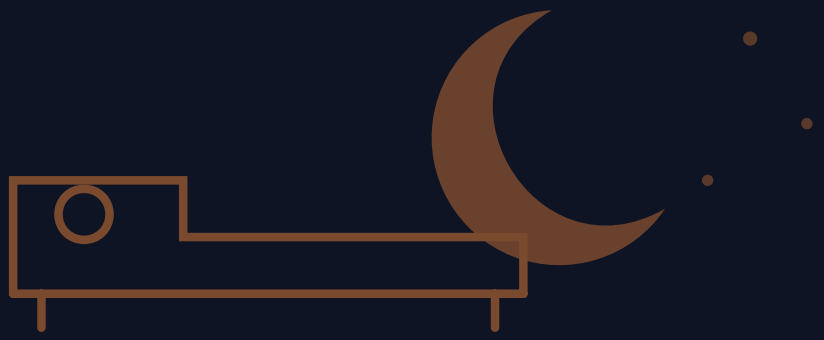
Ask your prescriber first, and treat the answer as the instruction, if any of these is true of you: a heart attack in the last twelve months, unstable chest pain, severe aortic stenosis, an arrhythmia, blood pressure that is not controlled, very low blood pressure, or pregnancy. **Cold specifically:** Raynaud's. **Heat specifically:** if you are a man trying to conceive, halve the dose rather than removing it.

**What we are standing on.** Sauna frequency and duration: *We are confident*, and the evidence is observational, which means it shows people who use a sauna more often do better and cannot prove the sauna is why. Cold exposure: *Early evidence*. The doses below are the protocol's, not a finding.

> **Answers Q1.** Controlled stress that builds mitochondrial capacity. Nothing on our panel isolates it.

# Sleep

Same time, every night. Dark, cool, nothing plugged in.



DARK, COOL, SAME TIME

6

06 / SLEEP

# Sleep

BEGINNER

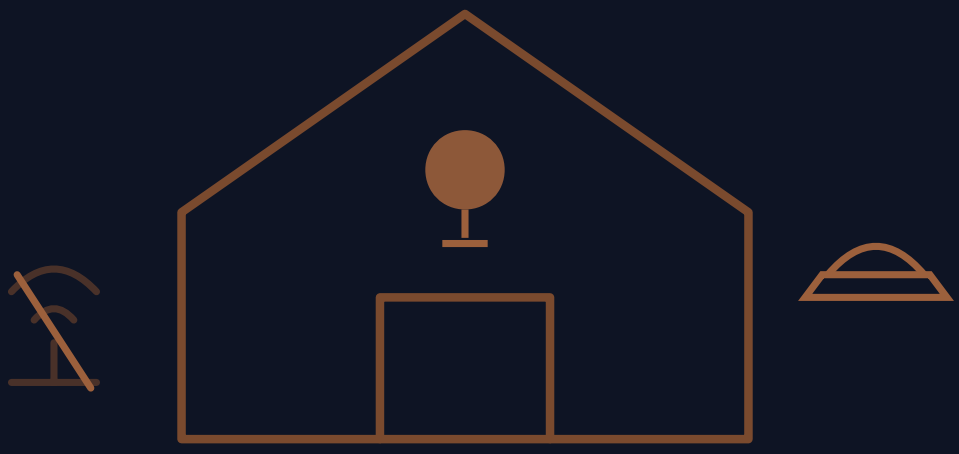
Same bedtime, same wake time, seven days a week. Room cool and completely dark. Last food three hours before bed. Phone out of the bedroom.

Pick a wake time and hold it. That is the only sleep rule in month one, and it is the one that makes everything else possible.

> **Answers Q3, Q4.** Recovery is when gradients are rebuilt and membranes repaired. Sleep timing and regularity are logged daily.

# Equipment and sourcing

Glasses, bulbs, a router timer, water, and where to get each one.



RED LIGHT, ROUTER OFF, GLASSES ON

7

07 / ENVIRONMENT

# Equipment and sourcing

BEGINNER

Dr. Pittman takes no commission on anything listed here. These are what he uses.

**Specifications, not brands.** Each row says what the thing has to do, so you can judge any product against it yourself. No brand is named in this table. **We take no commission on anything named in this guide, and nothing here is a paid placement.** Evidence tier for this whole table: **Dr. Micah's idea, being tested**, except the rows marked otherwise.

| | |
|---|---|
| Evening glasses | Lenses that block 460 to 490 nm. Ask any seller for the transmission spectrum and do not buy without it, because "blue light blocking" is unregulated and most amber lenses do not reach 490 nm. Red lenses block more than amber. *Tier: early evidence for the effect on melatonin onset.* |
| Screens | Any color-temperature shifter set to remove blue after sunset. The free ones work. A film on the monitor does the same job in hardware if you prefer not to run software. |
| Bulbs | Incandescent wherever you can still find them, because they have almost no blue. Otherwise a red or amber bulb in the rooms you use after dark. |
| Router | Mechanical outlet timer, off 10pm to 6am. The cheapest item on this list. |
| Grounding, optional | **Dr. Micah's idea, being tested.** Not a required part of the protocol and the evidence for it is weak. If you want to try it, a conductive sheet for the bed or a mat for the desk is the usual form. Test the outlet with an inexpensive outlet tester first. A participant who never grounds is fully adherent. |
| Water | Mineral water with a measurable mineral content, not distilled or reverse-osmosis alone. At home, under-sink reverse osmosis with salt added back, a quarter teaspoon per liter. *The specification for the salt: an unrefined salt with a full trace-mineral profile, not refined sodium chloride. Unrefined salts vary and most table salt does not qualify, so read the mineral analysis rather than the front of the packet.* |
| Sauna | Traditional, not infrared. A gym sauna is fine and is what most people should use. *Tier: strong for cardiovascular outcomes in the Finnish cohort data.* |
| Cold | A cold shower is enough to start. A chest freezer with a GFCI is the cheap home plunge. *Tier: contested. One unreplicated trial, and the warm-shower arm improved about as much.* |
| Butter | Grass-fed, any brand. |
| Testing | A dried-blood-spot Omega-3 Index kit at day 0 and day 90, from a lab that reports EPA+DHA as a percentage of red cell membrane fatty acids. A hand dynamometer for grip. *Tier: established for the Omega-3 Index as a defined measurement.* |

# Three phases

Stop the drain. Recharge. Build. Measured at both ends.




MEASURED AT BOTH ENDS

8

08 / THE NINETY DAYS

# Three phases

BEGINNER

| | |
|---|---|
| Days 1 to 30 | Light and water. Fifteen minutes outside in the morning, fifteen in the afternoon, fifteen at sunset. Glasses on at sundown, screens filtered, phone out of the bedroom. The morning glass and three liters. A fixed wake time. That is everything, and it is more than it sounds. |
| Days 31 to 60 | Movement and the window. Walk after dinner. 9am to 7pm window, no snacks. Two ten-minute resistance sessions. One new food a week. |
| Days 61 to 90 | Load. Three resistance days. Longer walks. Gluten and seed oils out. Cold finish on the shower. Day 90 draw. |

# Daily checklist

This is what goes in the log. Under a minute.



UNDER A MINUTE, EVERY DAY

9

09 / EVERY DAY

# Daily checklist

BEGINNER

## Morning daylight

- [ ] Outside within 30 min of waking, 15 min, barefoot
- [ ] Fifteen minutes of afternoon sun
- [ ] Fifteen minutes at sunset

## Hydration

- [ ] Morning glass
- [ ] Three liters of water

## Movement

- [ ] Walk after a meal

## Food timing

- [ ] Nothing three hours before bed

## Heat and cold

- [ ] Cold finish on shower, 30 to 60 s

## Sleep

- [ ] Same wake time

## Nighttime darkness

- [ ] Glasses on when the sun goes down
- [ ] Screens filtered
- [ ] Phone out of the bedroom
- [ ] Logged

THE HONEST PART

## What this is, and is not

The Human Battery Project is an educational wellness program, not medical treatment. It does not diagnose or treat any condition and does not replace your physician.

Any laboratory result outside the reference range is referred to a physician. Every time.

If you take prescription medication, review any change to your eating window, and any sauna or cold exposure, with your prescriber before starting. Never adjust a medication on your own because of anything in this document.

Stop and seek medical care for chest pain, fainting, shortness of breath at rest, or any new symptom that concerns you.

Never manipulate potassium or other electrolytes to try to change your body's charge. Balance and correction of a genuine deficiency only.

> **We do not promise outcomes.** We measure at both ends and show you exactly what changed.